Shamar Weekes

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EDUCATION

Florida International University **Expected Graduation: December 2026**
Bachelor of Science in Computer Science *Miami, FL*

- **Relevant Coursework:** Programming II, Data Structures, Systems Programming, Principles of Programming Languages, Software Engineering I, Computer Architecture, Technology in the Global Arena
- **Organizations:** CodeCrunch, INIT, BreakThrough Tech, Panther Robotics, Women in Computer Science (WiCS)

EXPERIENCE

Founder & Software Engineer **November 2025 – Present**
CareLog

- Built a healthcare operations platform (FastAPI, React, PostgreSQL, Docker, GCP) replacing paper care logs, currently piloting at a **40-bed** facility serving **25** rotating caregivers.
- Architected a multi-tenant backend with **30+ REST APIs**, role-based access control, and MQTT integration to process real-time events from proprietary hardware.
- Eliminated a critical race condition in care verification under concurrent access, validating the fix against **100 concurrent users** in Locust load tests.
- Secured healthcare data with field-level encryption, JWT auth, and audit logging, supporting **HIPAA**-aligned operational workflows.

Open Source Contributor **August 2025 – Present**
Kestra, TensorFlow

- **Delivered PR #7321 to Kestra (10k+ GitHub stars)**, fixing a frontend rendering defect affecting the platform's legend component.
- **Drove root-cause analysis across 3 pull request iterations**, isolating the root cause and refining the implementation based on maintainer feedback until successful upstream acceptance.
- **Collaborated with distributed open-source maintainers** using Git, pull requests, and code reviews in a large production codebase.
- **Analyzed the architecture of TensorFlow** to identify high-impact contribution opportunities and deepen understanding of large-scale ML systems.

PROJECTS

Real-Time Human Activity Recognition **March 2026 – Present**
Python, TensorFlow, OpenCV, Flask

- Built a real-time human activity recognition system using TensorFlow and OpenCV to classify actions like walking, running, sitting, and jumping with 92% accuracy on personal webcam data.
- Collected and annotated 1,200+ video clips from personal workouts and daily activities to train and validate the model.
- Designed a Python + Flask dashboard displaying live predictions, confidence scores, and activity duration in real-time.
- Optimized model with dropout, early stopping, and data augmentation, reducing misclassification by 28% and training time by 35%.

C/C++, Embedded Systems, Sensors, Microcontrollers **FIU-SEDS | Vensors | January 2025 – April 2025**

- Develop and integrate code for the Vensors project, controlling flight sensors and accelerometers on a microcontroller platform.
- Adjust and optimize full system setup through PRs (16, 25), reducing initialization errors by **90%** and ensuring accurate data collection across **3+ microcontrollers**.
- Automate calibration and validation, cutting setup and testing time from **2 hours to 15 minutes per session**, while achieving **100% measurement accuracy** over **50+ test cycles**.
- Collaborate with a team of **9+ engineers**, contributing to debugging and code integration, accelerating the overall development timeline by **30%**.

LEADERSHIP

Workshop Lead, CodeCrunch **September 2025 – December 2025**
Miami, FL / Zoom

- Led twice-weekly hybrid networking workshops for FIU students, with hands on practice and lecture slides.
- Developed all curriculum materials and exercises from scratch, to facilitate student learning and engagement.

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C, C++, C# , SQL, HTML, CSS
Frameworks & Libraries: React, Next.js, Node.js, Express, Django, Flask, FastAPI, Tailwind CSS
Databases: PostgreSQL, MongoDB, MySQL, SQLite, Microsoft SQL Server, Redis, Firebase
Tools & Technologies: Docker, Git, CI/CD, Linux, REST APIs, PyTest, JUnit, Vitest, Figma, Agile, OOP

CERTIFICATIONS

Microsoft Certified: Introduction to Computers and Operating Systems and Security **November 2023**